DATE: 23/01/26

EXP NO: 3

STUDY OF DIFFERENT TYPES OF NETWORK CABLES AND CRIMPING OF CABLE WITH RJ45 CONNECTOR

AIM:

To Study different types of network cables and understand the crimping process using an RJ45 Connector.

CONCEPT:

Network cables are used to connect Computers and other networking device for data communication. The most commonly used cable in LAN is the unshielded Twisted Pair (UTP) cable.

An RJ45 connector is attached to the Ends of the cable to Establish a network connection. The wires are arranged according to the T568A or T568B wiring Standard before crimping, and the connected cable is tested to ensure Proper Connectivity.

TYPE OF NETWORK CABLES:

CABLE TYPE	DESCRIPTION
1. UTP (unshielded Twisted Pair)	commonly used for Ethernet LAN's and office network
2. STP (Shielded Twisted Pair)	used in area with high electrical or Electromagnetic interference!
3. coaxial cable	used for cable television, broadband internet, and CCTV systems.

CRIMPING PROCEDURE:

1. cut the UTP cable to the required length
2. Strip about 2-3 cm of the outer insulation
3. untwist and straighten the wires
4. Arrange the wires according to the 568B Standard
5. Trim the wire end evenly.
6. Insert the wires into the RJ45 connector

OBSERVATION:

* The wires were arranged according to the T568B standard.
* The RJ45 connector was crimped successfully.
* The cable tester indicated proper continuity.
* The cable was ready for network communication.

RESULT:

Thus different types of network cables were studied and the UTP cable was successfully terminated using RJ45 connectors and tested for proper connectivity.